

Monica Joya

ITAI 3377: Artificial Intelligence for Edge and IoT Devices

A03: Case Study Analysis on Edge-Computing Video Analytics...

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Case Study Analysis on Edge-Computing Video Analytics for Real-Time Traffic Monitoring in a Smart City

A. Introduction and Objectives

In early 2018, the Australian federal government funded a joint effort between a local council and university researchers in Liverpool, New South Wales—a rapidly expanding Sydney suburb of about 27,000 people facing a CBD overhaul projected to draw tens of thousands of new daily visitors (Barthélemy et al., 2019). The research team set out to develop a compact, AI-powered sensor that could connect to cameras the city already owned, identify different types of road users as they moved through the area, and do so entirely on-device so that no identifiable images would ever be stored or sent over a network (Barthélemy et al., 2019).

A related issue is that local governments across Australia pour significant resources into surveillance camera systems, yet these networks serve almost no purpose beyond reviewing security incidents (Wilson & Sutton, 2003). Strict privacy laws limit access to the footage, which means valuable movement data sits unused. At the same time, global urbanization trends suggest that the majority of people worldwide will soon reside in cities (United Nations, 2017), making smarter approaches to monitoring and managing foot and vehicle traffic increasingly urgent. This project aimed to turn underused CCTV infrastructure into a real-time urban planning resource by processing video locally and sending only anonymized metadata to planners.

B. Methodology

The team began by holding two community workshops (35 total participants) with residents and city planners. Four requirements emerged: the sensor had to detect multiple road user types (pedestrians, vehicles, cyclists), be fully privacy-compliant with no personal data stored or shared, leverage the city's existing CCTV cameras, and be scalable so new sensors could be added over time regardless of their technology (Barthélemy et al., 2019).

Development followed a phased approach: build a prototype, validate accuracy on a standardized benchmark dataset, measure system resource usage during a live CCTV connection, then deploy 20 sensors citywide for field testing. Design constraints included a 35W power budget, an IP67-rated weatherproof enclosure for outdoor use, compatibility with IP cameras, and support for LoRaWAN wireless communication—which imposed a minimum 5-minute interval between data transmissions (Barthélemy et al., 2019).

C. Technology and Implementation

Hardware

At the center of the device is an NVIDIA Jetson TX2—a compact, energy-efficient computing board equipped with a 256-core graphics processor built to handle deep learning

workloads, along with 8 GB of RAM and 32 GB of onboard storage, all operating on roughly 15 watts of power (Barthélemy et al., 2019). It is paired with a Pycom LoPy 4 module for LoRaWAN wireless communication, which allows long-range, low-power data transmission in locations without internet access. The unit sits inside a waterproof aluminum heat-sink case. For larger deployments, the team also built a rack version holding 15 independent Jetson TX2 modules in a single chassis, where each module processes one camera's feed and operates independently so a single failure does not affect the others (Barthélemy et al., 2019).

Software

Detection uses YOLO V3 (“You Only Look Once”), a deep neural network that processes an entire image in a single pass rather than scanning it multiple times. It detects objects at three size scales—important for CCTV footage where objects appear larger or smaller depending on distance from the camera—and was configured to recognize pedestrians, bicycles, cars, buses, trucks, and motorbikes. The implementation ran in PyTorch on the Jetson's GPU via CUDA (Barthélemy et al., 2019; Redmon & Farhadi, 2018).

Tracking uses the SORT algorithm (“Simple Online and Real-time Tracking”), which predicts where each object should appear in the next frame using a Kalman filter, then matches predictions to actual detections using the Hungarian algorithm. Each tracked object gets a unique ID to record its path. However, if someone leaves and re-enters the camera's view, they receive a new ID—the system cannot reconnect them to their previous track (Barthélemy et al., 2019; Bewley et al., 2016).

Edge-Computing Paradigm

All processing happens on the device itself. No raw video is transmitted—only metadata like counts and trajectories. This keeps bandwidth low and ensures privacy since no images leave the CCTV network (Satyanarayanan, 2017; Shi et al., 2016). This aligns with the edge computing principles from Module 2, where local processing at the network's edge reduces dependence on centralized cloud infrastructure.

D. Validation and Performance

The team validated accuracy on the Oxford Town Center Dataset—a 1920×1080, 25 fps video of a busy street with 230 annotated pedestrians across 4,500 frames. According to the results, the sensor demonstrated a “mean accuracy of 69% with a median relative error of 33%” (Barthélemy et al., 2019, p. 9). It consistently undercounted rather than overcounted, mainly because when people stood close together, the algorithm's non-maximum suppression step sometimes merged overlapping detection boxes into one. Accuracy was higher with fewer people and dropped as crowds got denser (Barthélemy et al., 2019).

Processing speed averaged about 19.6 fps with a max of ~23 fps. Speed and detection count were inversely related: more objects meant fewer frames processed per second. The bottleneck was the SORT tracker running on the CPU rather than the GPU, so it could not leverage the Jetson's CUDA cores. System monitoring during a live CCTV test confirmed stable memory, disk, and temperature, and that outgoing data was far smaller than incoming video—validating the bandwidth efficiency of edge processing (Barthélemy et al., 2019).

E. Real-World Applications

Indoor: Emergency Evacuation

For the initial field test, the research team installed the device on the first floor of a university facility at the University of Wollongong, pointed at the main stairway entrance. During the one-hour monitoring session, a real fire alarm was triggered unrelated to the study (Barthélemy et al., 2019). What made this interesting is that the sensor data clearly captured the event: activity dropped to zero during the alarm, confirming everyone had left the floor, and a smaller wave of people returned once it was declared safe. In total, the device logged 631 tracked individuals, and the resulting movement paths and heat maps lined up with what you would expect for that layout. The authors pointed to this as evidence that the sensor could be a useful tool for spotting unusual crowd behavior during emergency situations (Barthélemy et al., 2019).

Outdoor: Liverpool City Center

Twenty sensors were deployed citywide. Data from one camera near a busy pedestrian street over one week (February 20–27, 2019) showed clear daily rhythms: pedestrian activity peaked between 8:00 AM and 4:00 PM, vehicles showed two peaks around 9:00 AM and 4:00 PM (commute hours), and Sunday was the quietest day. On February 23 alone, the sensor logged 20,399 unique objects. Trajectory visualizations revealed two dominant pedestrian flow corridors through the intersection (Barthélemy et al., 2019).

The data collection infrastructure used the Agnosticity framework, built on the open-source OM2M platform implementing the oneM2M interoperability standard. This allowed visual sensors, air quality monitors, and noise sensors to feed data into a common platform with RESTful API access and a web dashboard—connecting to the sensor network and edge gateway concepts from Module 3 (Barthélemy et al., 2019).

F. Challenges and Future Work

Challenges

Key limitations included the 69% accuracy (driven by occlusion in crowds), the CPU-bound tracking algorithm creating a speed bottleneck, LoRaWAN's 5-minute minimum transmission interval, and the inability to re-identify objects that left and re-entered the camera view (Barthélemy et al., 2019).

Authors' Proposed Improvements

The authors recommended upgrading to the NVIDIA Xavier platform (double the CUDA cores, preliminary tests showed 2x speed improvement), porting YOLO V3 from PyTorch to optimized frameworks like TensorFlow with TensorRT, and implementing the SORT tracker on the GPU to eliminate the CPU bottleneck (Barthélemy et al., 2019).

Developments Since 2019

Since publication, relevant technology has advanced significantly. NVIDIA's Jetson Orin series (2022) delivers dramatically more processing power in the same low-power form factor. The YOLO family has progressed through versions 4–11 with better accuracy, speed, and built-in optimization for edge hardware. Trackers like DeepSORT, ByteTrack, and BoT-SORT have

improved occlusion handling by incorporating appearance features—directly addressing this study’s main accuracy weakness. Inference optimization tools like TensorRT and ONNX Runtime have matured, and 5G networks offer better bandwidth for IoT deployments.

G. Personal Evaluation

The project’s greatest strength is its privacy-by-design approach. Processing everything on-device and transmitting only anonymous metadata avoids the ethical and legal complications of sending video footage across networks. The community-first methodology—holding workshops before building anything—meant the design was driven by actual needs rather than assumptions.

The main weakness is that 69% accuracy means consistent undercounting, especially during the crowded peak hours when precise data matters most. The one-week outdoor deployment, while informative, is too short for the seasonal and long-term trend analysis that urban planners need.

Beyond what the authors proposed, I would suggest adopting a modern tracker like DeepSORT to reduce identity-switching errors, integrating the traffic data with the co-located air quality and noise sensor data through the Agnosticity platform for richer analysis, and adding automated anomaly detection to turn the system from passive monitoring into active alerting for crowd safety events. Overall, the project is a solid proof of concept that demonstrates that edge AI and existing CCTV infrastructure can work together as practical urban planning tools and, with today’s hardware and software improvements, a similar deployment could achieve substantially better results.

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